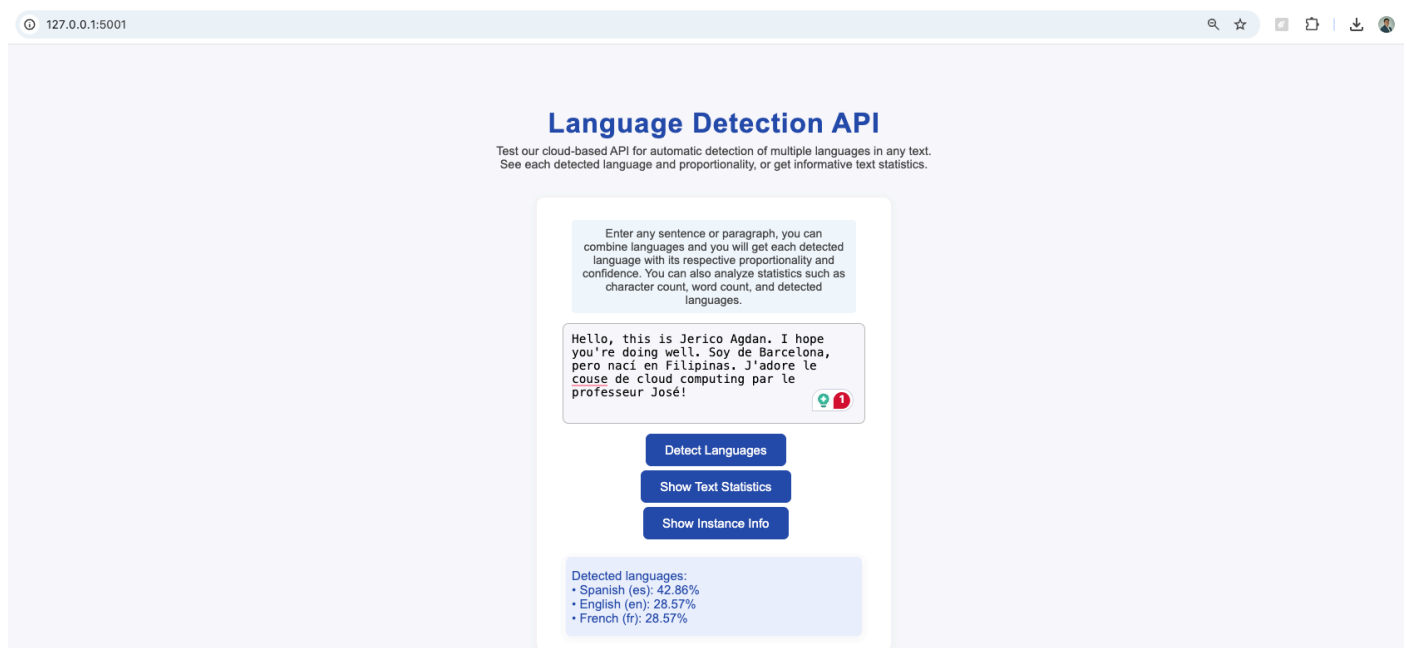


Exercise 1

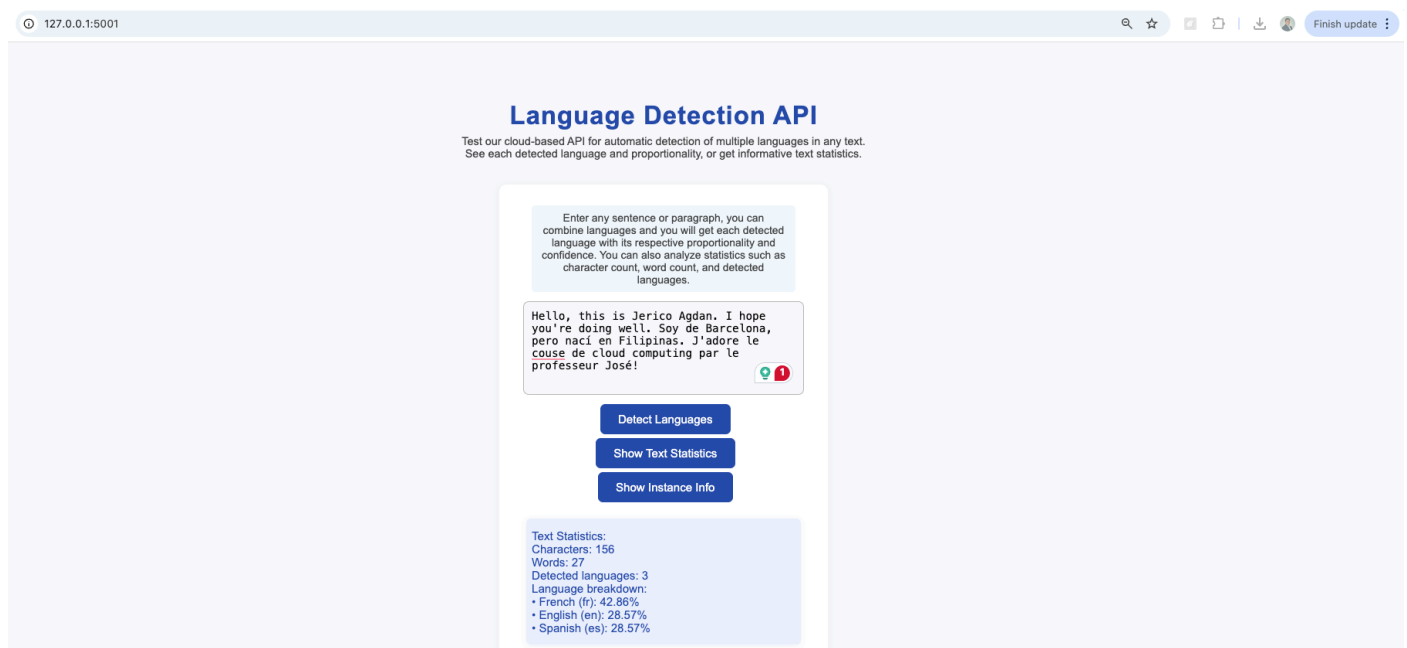
You can find the Flask code and HTML in my GitHub account (<https://github.com/JericoNeil/Cloud-Computing-Flask-Assignment.git>). The **Read Me** file explain step by step how I created the API regarding the language detection. It is available specifically in the “**Assignment1_LanguageDetect**” folder.

Below you'll find some screenshots:

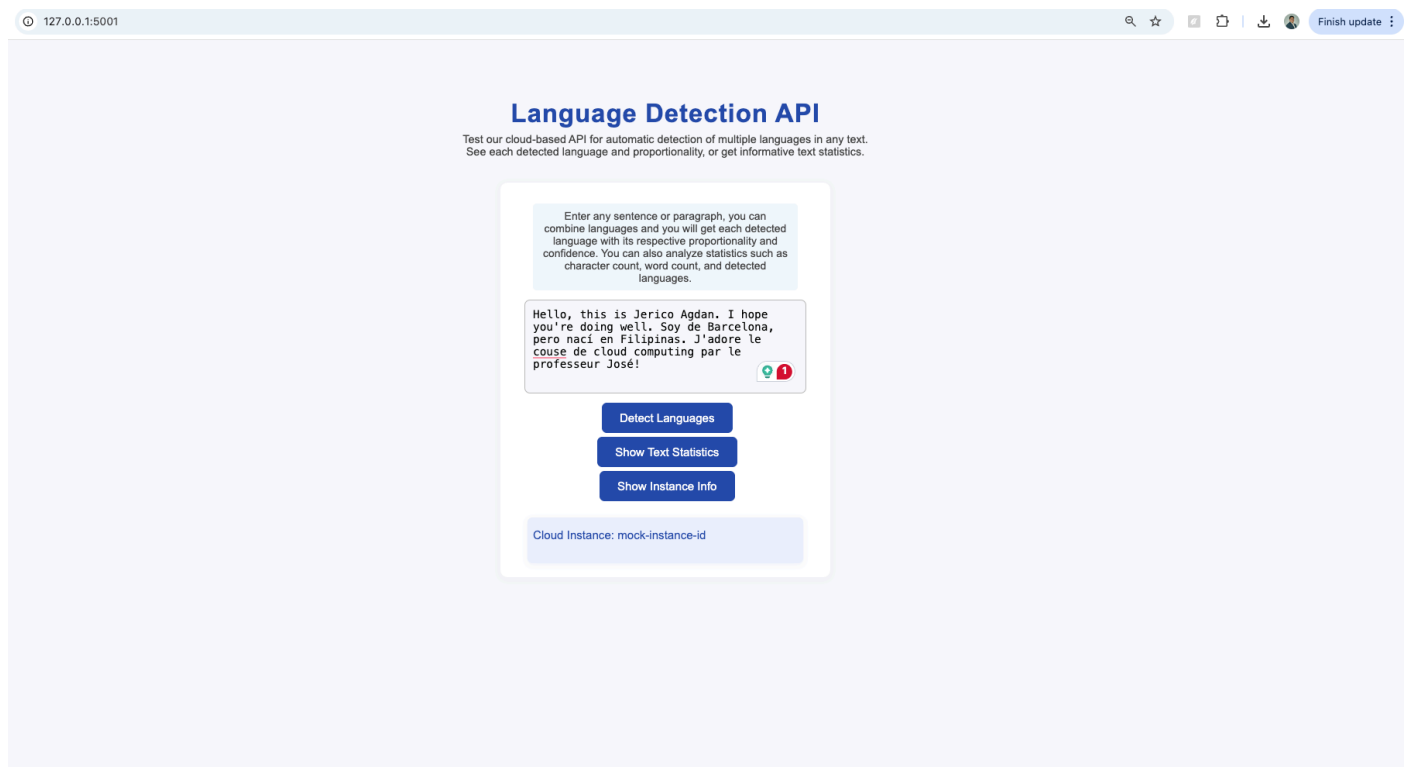
This first screenshot is about the **/detect-all-probabilities endpoint**, which shows the detected languages and the probabilities of each respective language in the text.



The following screenshot is about the **/text-stats endpoint**, which returns summary statistics about input text including the number of characters, words, detected languages and languages breakdown:



Thirdly, this is **/instance** endpoint that will be used for the cloud compute instance information. As it's not launched with an EC2 instance yet, it returns a mock instance id.

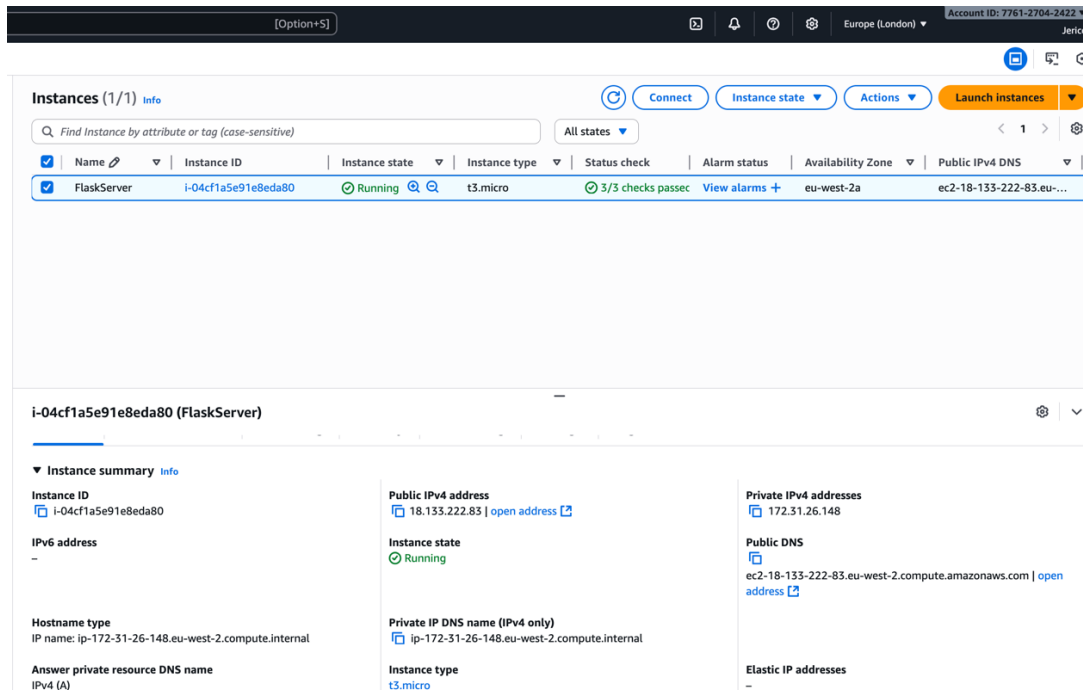


And finally, this is a screenshot regarding the terminal showing that the API works perfectly with all endpoints:

```
❯ (langdetect_env) jericoagdan@Jericos-MacBook-Air Exercise1_LanguageDetect % python language_api.py
* Serving Flask app 'language_api'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5001
* Running on http://192.168.0.43:5001
Press CTRL+C to quit
127.0.0.1 - - [12/Nov/2025 21:54:37] "POST /detect-all-probabilities HTTP/1.1" 200 -
127.0.0.1 - - [12/Nov/2025 21:54:37] "POST /text-stats HTTP/1.1" 200 -
127.0.0.1 - - [12/Nov/2025 21:54:38] "GET /instance HTTP/1.1" 200 -
127.0.0.1 - - [12/Nov/2025 21:55:26] "POST /text-stats HTTP/1.1" 200 -
127.0.0.1 - - [12/Nov/2025 21:55:26] "POST /detect-all-probabilities HTTP/1.1" 200 -
127.0.0.1 - - [12/Nov/2025 21:55:37] "POST /detect-all-probabilities HTTP/1.1" 200 -
127.0.0.1 - - [12/Nov/2025 21:55:49] "POST /detect-all-probabilities HTTP/1.1" 200 -
127.0.0.1 - - [12/Nov/2025 21:56:10] "POST /text-stats HTTP/1.1" 200 -
```

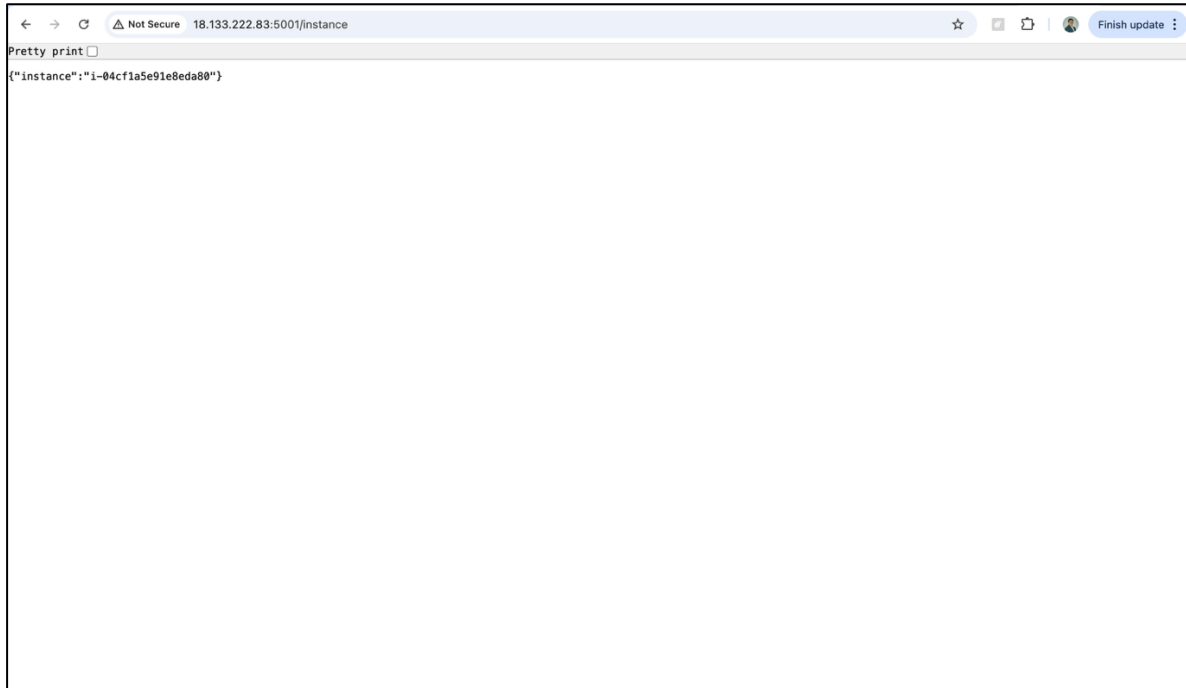
1) Screenshot of the running instance

Screenshot of AWS's instances with the list of instances, with the instance in "Running" state, the public IP and username at the top right:

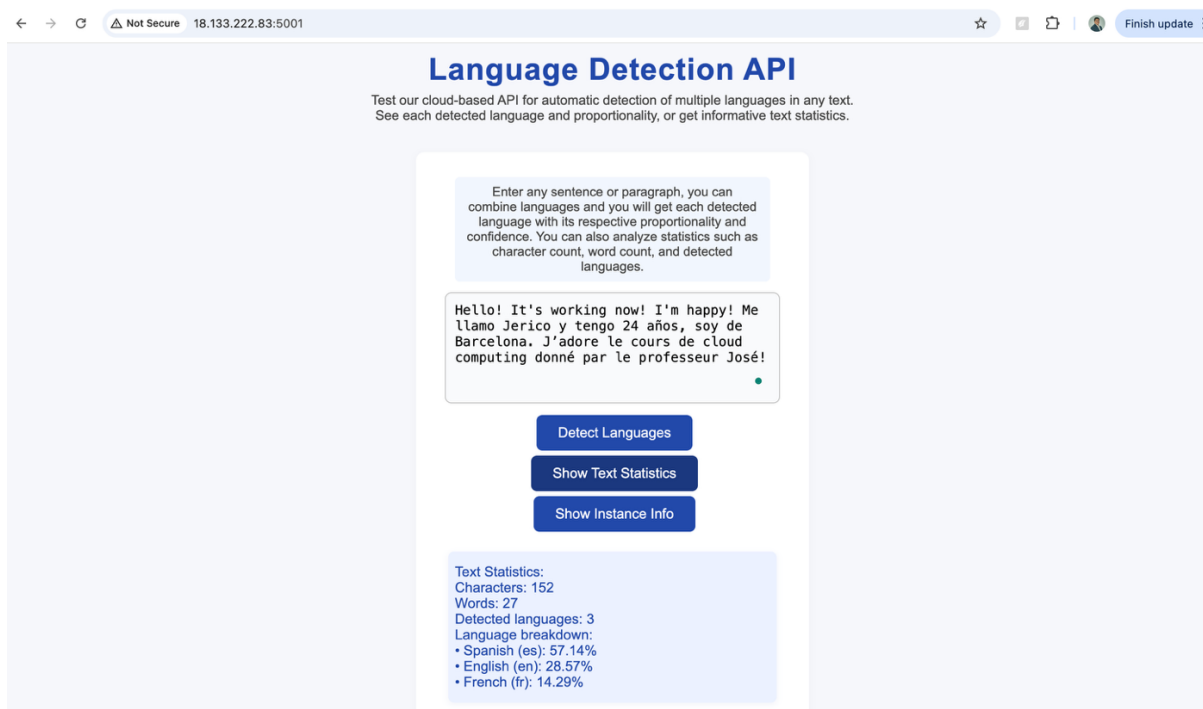


2) Screenshot of the request to the /instance endpoint, showing that the API works with the EC2 instance





3) Extra: Screenshot of the API working with the Text Statistics endpoint



It shows the number of characters, words, detected languages and language breakdown with its respective proportion.

Exercise 3

You can find this exercise in my GitHub account: <https://github.com/JericoNeil/Cloud-Computing-Flask-Assignment.git>

The **Read Me** file explains more about the project. All materials for this exercise are also located in my GitHub account, specifically in the “**Assignment3_CareerPath**” folder.

1. INTRODUCTION OF THE PROBLEM

For this exercise, I developed a recommender system that enables high school students to type as input their Bachillerato and PAU grades and discover a tailored set of university degrees that they may realistically pursue based on their predicted or actual scores.

The motivation behind this project is personal, I want to help my brother decide which university degree he can aim for. He is currently in his first year of high school and, like many students, is unsure about what bachelor’s degree to choose. Some people, like my brother, would be open to study from a BBA degree to Economics and even other degrees in the field of social sciences. Despite the significance of this decision, there is no existing platform that shows students the full spectrum of career paths available based on their predicted or actual high school and PAU grades. For instance, in regards to the cut-off grades in Catalonia for bachelor degrees, you can find them in the website from the Government of Catalonia, but the information of studies is in PDF, which is not very convenient to filter degrees: <https://universitats.gencat.cat/ca/preinscripcions/notes-tall/>



Notes de tall 1a assignació Juny 2025 (11/07/25)						
Ordre alfabètic per centre d'estudi						
ESTUDIS DE GRAU I TÍTOLS PROPIS						
Codi	Nom del centre d'estudi i població	Sigles Universitat	PAU / CFGS	Més grans de 25 anys	Titulats universitaris	Més grans de 45 anys
41029	Administració d'Empreses i Gestió de la Innovació "Tecnocampus" (Mataró)	UPF	7,688	5,000	5,000	5,000
41036	Administració d'Empreses i Gestió de la Innovació "Tecnocampus" (docència en Anglès) (Mataró)	UPF	7,628	5,000	5,000	5,000
41041	Administració d'Empreses i Gestió de la Innovació / Màrqueting i Comunitats Digitals "Tecnocampus" (simultaneïtat) (Mataró)	UPF	7,546	5,000	5,000	5,000
41035	Administració d'Empreses i Gestió de la Innovació / Turisme i Gestió del Lleure "Tecnocampus" (simultaneïtat) (Mataró)	UPF	5,000	5,000	5,000	5,000
11033	Administració i Direcció d'Empreses (Barcelona)	UB	9,348	5,000	5,000	5,000
41008	Administració i Direcció d'Empreses (Barcelona)	UPF	10,966	5,000	5,000	5,000
71063	Administració i Direcció d'Empreses "IGEMA" (Barcelona)	URV	5,000	5,000	5,000	5,000
21014	Administració i Direcció d'Empreses (Cerdanyola del Vallès)	UAB	9,804	5,000	5,000	5,000
21099	Administració i Direcció d'Empreses (docència en Anglès) (Cerdanyola del Vallès)	UAB	10,128	5,000	5,000	5,000
81003	Administració i Direcció d'Empreses (Girona)	UdG	9,218	5,000	5,000	6,333
61069	Administració i Direcció d'Empreses (Igualada)	UdL	8,224	5,000	5,000	5,000
61003	Administració i Direcció d'Empreses (Lleida)	UdL	7,764	5,000	5,000	5,000
91042	Administració i Direcció d'Empreses (Campus Manresa) (Manresa)	UVic-UCC	5,716	5,000	5,000	5,000
71002	Administració i Direcció d'Empreses (Reus)	URV	8,413	5,000	5,000	5,000
71004	Administració i Direcció d'Empreses (Tortosa)	URV	7,220	5,000	5,000	5,000
91003	Administració i Direcció d'Empreses (Vic)	UVic-UCC	6,420	5,000	5,000	5,000
11064	Administració i Direcció d'Empreses / Dret (simultaneïtat) (Barcelona)	UB	10,692	5,000	6,860	5,000

Moreover, you also have this platform from notasdecorte.es (<https://notasdecorte.es/zona/cataluna>), in which you can have access to all degrees not only from Catalonia. This tool offers filters to select a specific university or to search for a particular degree if the student already knows what they are looking for. However, many students especially those like my brother who are still exploring their options, may not yet have a clear target in mind and would benefit from a broader, more exploratory tool.

Notas de corte: Cataluña 2024-2025

La nota de corte más alta de Cataluña en el curso 2024-2025 fue de un 13,111 para estudiar el Doble Grado en Física + Matemáticas en la Universitat Autònoma de Barcelona

Abajo se muestran datos de los 856 grados ofrecidos en Cataluña, de los que 506 se imparten en centros públicos, y 350 en centros privados

Importante: Recuerda que es imposible saber de antemano qué nota de acceso tendrás que sacar para entrar en cualquier carrera de Cataluña este año. Las notas de corte del año pasado son sólo orientativas, ya que cambian cada año en función de la demanda y del número de plazas ofrecidas

Filtrar por universidad

Barcelona Culinary Hub
Bau - Centro Universitario de Diseño de Barcelona
CESDA (Centro de Estudios Superiores de la Aviación)
CETT - Campus de Turismo, Hotelería y Gastronomía
Campus Docent Sant Joan de Déu
Centro de la Imagen y la Tecnología Multimedia (CITM)
E.U. de Enfermería del Hospital de la Santa Creu i Santa Pau
E.U. de Hotelería y Turismo Sant Pol de Mar
E.U. de Relaciones Laborales de Lleida
E.U. de Turismo Euroaula
E.U. de Turismo y Dirección Hotelera
E.U. de la Salud y el Deporte (URV)
E.U. de la Salud y el Deporte (UdG)
EAE Business School
EINA, Centro Universitario de Diseño y Arte de Barcelona (UAB)
ELISAVA Escuela Universitaria de Diseño e Ingeniería de Barcelona
ESERP Nístel Business & Law School

Titulaciones

Doble Grado en Física + Matemáticas
Universitat Autònoma de Barcelona

Universidad Pública
Web de la facultad: <http://www.uab.cat/ciencias>
Duración: 5,0 años
Precio del primer curso: 1.108 €

¡Pídeles información ¡GRATIS!

Doble Grado en Física + Matemáticas
Universitat de Barcelona

Universidad Pública
Web de la facultad: <http://www.mat.ub.es>
Duración: 5,0 años
Precio del primer curso: 1.181 €

¡Pídeles información ¡GRATIS!

Barcelona
Presencial

Nota de corte
13,111

Idioma de enseñanza:
Bilingüe
(castellano/lengua
cooficial)

Barcelona
Presencial

Nota de corte
12,864

Idioma de enseñanza:
Lengua cooficial

Hence, my project provides a more intuitive and user-focused experience by enabling students to discover degree programs that truly fit their academic profile and interests, even if they don't have a predetermined major or university in mind. The aim is to empower high school students to make more informed and personalized choices about their academic and career paths.

2. DATA CLEANING AND STRUCTURING

In order to carry out the project, I used the Database from the Generalitat of Catalonia website, which is the official website to retrieve the "cut-off grades" or "Notes de Tall" (in Catalan), of each degree of Catalanian universities as mentioned before (excluding private university bachelor's). Unfortunately, we do not have the Excel/CSV File, so I converted the PDF file to Excel and followed a multi-step data cleaning process.

Notes de tall 1a assignació Juny 2025 (11/07/25)						
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ESTUDIS DE GRAU I TÍTOLS PROPIS						
Codi	Nom del centre d'estudi i població	Sigles Universitat	PAU / CFGS	Més grans de 25 anys	Titulats universitaris	Més grans de 45 anys
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41036	Administració d'Empreses i Gestió de la Innovació "Tecnocampus" (docència en Anglès) (Mataró)	UPF	7.628	5.000	5.000	5.000
41041	Administració d'Empreses i Gestió de la Innovació / Màrqueting i Comunitats Digitals "Tecnocampus" (simultaneïtat) (Mataró)	UPF	7.546	5.000	5.000	5.000
41035	Administració d'Empreses i Gestió de la Innovació / Turisme i Gestió del Lleure "Tecnocampus" (simultaneïtat) (Mataró)	UPF	5.000	5.000	5.000	5.000
11033	Administració i Direcció d'Empreses (Barcelona)	UB	9.348	5.000	5.000	5.000
41008	Administració i Direcció d'Empreses (Barcelona)	UPF	10.966	5.000	5.000	5.000
71063	Administració i Direcció d'Empreses "IGEMA" (Barcelona)	URV	5.000	5.000	5.000	5.000
21014	Administració i Direcció d'Empreses (Cerdanyola del Vallès)	UAB	9.804	5.000	5.000	5.000
21099	Administració i Direcció d'Empreses (docència en Anglès) (Cerdanyola del Vallès)	UAB	10.128	5.000	5.000	5.000

To do so, I also enriched the dataset by adding several **new columns (in red)** to be able to filter as can be seen below:

- **Location:** The city where each university program is offered
- **Province:** The province in which the university is located (for broader geographic filtering, there are 4 provinces in Catalonia)

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- **Field:** The academic or professional field of each degree (such as Health Sciences, Engineering, etc.)
- **High-School Specialization:** The typical Spanish Bachillerato track granting access to each degree (e.g., Science and Technology, Social Sciences, Arts and Humanities)

Notes de tall 1a assignació Juny 2025 (11/07/25)							
Ordre alfabètic per centre d'estudi							
ESTUDIS DE GRAU I TÍTOLS PROPIS							
Code	Degree	University	Location	Province	Field	High-School Specialization	PAU
21169	Física / Matemàtiques (simultaneïtat) (Cerdanyola del Vallès)	UAB	Cerdanyola del Vallès	Barcelona	Double Degree	Science and Technology	13.370
11020	Física / Matemàtiques (simultaneïtat) (Barcelona)	UB	Barcelona	Barcelona	Double Degree	Science and Technology	13.170
31112	Enginyeria en Tecnologies Aeroespacials, Grau en + Màster Universitari en Enginyeria Aeronàutica (PARS: Enginyer/a Aeronàutic/a)	UPC	Terrassa	Barcelona	Engineering and Architecture	Science and Technology	12.936
11016	Enginyeria Informàtica / Matemàtiques (simultaneïtat) (Barcelona)	UB	Barcelona	Barcelona	Double Degree	Science and Technology	12.898
11043	Medicina (Campus Clínic) (Barcelona)	UB	Barcelona	Barcelona	Health Sciences	Science and Technology	12.850
11059	Enginyeria Biomèdica (Barcelona)	UB	Barcelona	Barcelona	Engineering and Architecture	Science and Technology	12.740
21145	Dret / Relacions Internacionals (simultaneïtat) (Cerdanyola del Vallès)	UAB	Cerdanyola del Vallès	Barcelona	Double Degree	Social Sciences	12.714
41058	Medicina (Barcelona)	UPF	Barcelona	Barcelona	Health Sciences	Science and Technology	12.650
11083	Medicina (Campus Bellvitge) (L'Hospitalet de Llobregat)	UB	L'Hospitalet de Llobregat	Barcelona	Health Sciences	Science and Technology	12.650
31040	Matemàtiques (Barcelona)	UPC	Barcelona	Barcelona	Sciences	Science and Technology	12.644

In order to carry out the cleaning and structuring process, I did the following:

1. Extracting Location from Degree Name

All degree names in the dataset ended with their location in parentheses, for example, Administració i Direcció d'Empreses (Barcelona). My goal was to systematically extract just the place name (e.g., Barcelona).

To do this, I used two Excel formulas:

- **TEXTAFTER(A1, "(", -1):** This extracts all text after the last opening parenthesis in the string from cell A1. For the example above, it would yield Barcelona).
- **TEXTBEFORE(..., ")"):** Applying this to the previous result, it pulls everything before the closing parenthesis, giving just Barcelona.

Combined into one formula: `=TEXTBEFORE(TEXTAFTER(A1,"(", -1), ")")`.

fx: =TEXTBEFORE(TEXTAFTER(C6,"(", -1), ")")		
Degree	University	Location
Física / Matemàtiques (simultaneïtat) (Cerdanyola del Vallès)	UAB	Cerdanyola del Vallès
Física / Matemàtiques (simultaneïtat) (Barcelona)	UB	Barcelona
Enginyeria en Tecnologies Aeroespacials, Grau en + Màster Universitari en Enginyeria Aeronàutica (PARS: Enginyer/a Aeronàutic/a)	UPC	Terrassa

2. Creating and Filling the Region Column

Once I extracted the city names, I wanted to have a broader region classification, in this case, the **Catalonia provinces**: Barcelona, Tarragona, Girona and Lleida. This classification will be used to the final API so that students living in Girona, for example, can filter Girona-based universities.

In order to do so, I created an Excel tab called "Data_Cleaning" where I created a mapping table with two columns: the first one is the city values where I extracted these by executing the "Unique" Excel formula (**45 locations**) of the Column "Location" of the main tab, later I allocated the specific province (Barcelona, Girona, Tarragona, Lleida) of each city/region.

After that, I went back to the main tab and added the Region column by executing a **lookup** to fill the region column automatically for all rows. The photo on the right shows just some of the regions.

Unique Regions from Table 1 - Column E	Province
Mataró	Barcelona
Barcelona	Barcelona
Cerdanyola del Vallès	Barcelona
Girona	Girona
Igualada	Barcelona
Lleida	Lleida
Manresa	Barcelona
Reus	Tarragona
Tortosa	Tarragona
Vic	Barcelona
Tarragona / Reus	Tarragona
Reus / Barcelona	Tarragona
Tortosa / Barcelona	Tarragona
Tarragona	Tarragona
Sant Cugat del Vallès	Barcelona
Salt	Girona

3. Adding the Field + High School Specialization columns

To enable more advanced filtering and personalized recommendations, I enriched the dataset by adding two key columns: **Field** (field of the study of the bachelor's degree) and **High School Specialization** (the Bachillerato track that typically grants access to each degree). This allows the platform to let students filter degrees by either their current high specialization (Sciences and Tech, Arts and Humanities or Social Sciences), their field of interest of study, or both. For example, a student from a Science and Technology bachillerato could still explore management or law degrees if he/she finds them more suitable using the filter.

For the **Field** column, as there are 555 bachelor degrees, my goal was to assign every degree to a **broader academic area** such as "Health Sciences", "Engineering and Architecture" or, for combined programs, "Double Degree" (so I could provide a filter specifically for dual programs). To automate this field column, I used an LLM in which I prompted to extract relevant keywords from the degree title and map them to an academic area. For example, any degree containing "Enginyeria" was automatically mapped to "Engineering and Architecture." Wherever the lookup didn't retrieve a clear result, such as less common or ambiguous degrees (e.g., Pedagogia, Treball Social, etc.), I assigned the field manually to ensure accuracy. This process can be located in the "Data_Cleaning" Excel tab. The following screenshot is a non-exhaustive list of keywords:

Keyword	Field
(simultaneïtat)	Double Degree
Treball	Social Sciences and Law
Enginyeria	Engineering and Architecture
Arquitectura	Engineering and Architecture
Química	Sciences
Biologia	Sciences
Ciències Ambientals	Sciences
Matemàtiques	Sciences
Física	Sciences
Empresa	Social Sciences and Law
Administració	Social Sciences and Law
Economia	Social Sciences and Law
Dret	Social Sciences and Law
Turisme	Social Sciences and Law

For the **High School Specialization** column, I followed a similar approach. I built a mapping table that links each field to its most common Bachillerato specialization.

Batxillerat	Batxillerat
Double Degree	Manual
Social Sciences and Law	Social Sciences
Engineering and Architecture	Science and Technology
Sciences	Science and Technology
Health Sciences	Science and Technology
Arts and Humanities	Arts and Humanities

4. Final Cleaning & Export

As a final step, I made sure all categorical values were consistent (for example, always spelling "Barcelona" the same way), and ensured that all necessary fields were filled in. With all columns complete and clean, I then converted the Excel table into a JSON file, so it's much convenient for direct use in the API with the HTML backend.

3. ENDPOINTS and OUTPUT

For this project I used 3 endpoints:

- **/api/v1/students/calculate-score (POST)**: Calculates the student's university admission score (out of 14) based on Bachillerato, PAU exams, and specific subject marks.
- **/api/v1/students/recommendations (POST)**: Returns a ranked list of university degrees that best match the student's score, specialization, field of interest, location, and double degree preference.
- **/api/v1/statistics/overview (GET)**: Presents summary statistics including score distribution, fields, specializations, and universities in the dataset.

This is the initial view of the interface, where students can enter their Bachillerato and PAU grades, as well as specific subject scores, to begin the recommendation process.

The screenshot shows a web browser window with the URL 127.0.0.1:5002. The page has a purple header with the 'CareerPath' logo and the tagline 'Find your perfect university degree match based on your Bachillerato grades'. The main content area is titled 'Calculate Your Target Admission Score' and contains the following form fields:

- Bachillerato Grade (0-10):** A text input field with the placeholder 'e.g., 8.5'.
- PAU Exam Grades (5 exams, 0-10 each):** Five input fields for 'Spanish', 'Catalan', 'History', 'Foreign Language', and 'Elective'.
- Specific Subject Grades (at least 2, 0-10 each):** Four input fields for 'Subject 1', 'Subject 2', 'Subject 3 (optional)', and 'Subject 4 (optional)'.
- Calculate My Score**: A purple button at the bottom of the form.

Once grades are entered and the score is calculated, the interface displays the student's target admission score. This output corresponds to the functionality provided by the first API endpoint (calculate-score).

This screenshot shows the same 'Calculate Your Target Admission Score' form, but with data entered and the result displayed. The 'Bachillerato Grade' is 9. The 'PAU Exam Grades' are 10, 8.39, 8, 8.32, and 9. The 'Specific Subject Grades' are 10, 8, 4, and 4. The 'Calculate My Score' button is still present. Below the form, a large purple box displays the results:

Your Target Admission Score
12.497 / 14
Bachillerato: 5.4 | PAU: 3.497 | Specific: 3.6

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Next, students can set various filters according to their **field preferences and high school specialization** as well as other variables such as field of study, location, or double degree options. This process is linked to the second API endpoint (recommendations).



Get Degree Recommendations

Your Total Target Score (0-14):

Which filter do you want to use?

Your Bachillerato Specialization:

Field of Interest:

Preferred Province:

☒ Include Double Degrees

Find My Matches

Once you click Find My Matches, it will show the student the **Top 20 options**. The example shown below is a sample subset for demonstration purposes.

Found 36 matching programs. Here are your top 20:

- 1. Ciències Polítiques i de l'Administració / Dret (simultaneïtat) (Barcelona)** Safety

Double Degree

University: UB Location: Barcelona Field: Double Degree Cut-off Score: 11.475

Specialization: Social Sciences
- 2. Economia / Matemàtica Computacional i Anàlisi de Dades (simultaneïtat) (Cerdanyola del Vallès)** Safety Double Degree

University: UAB Location: Cerdanyola del Vallès Field: Double Degree Cut-off Score: 11.402

Specialization: Social Sciences
- 3. Ciència Política i Gestió Pública / Dret (simultaneïtat) (Cerdanyola del Vallès)** Safety

Double Degree

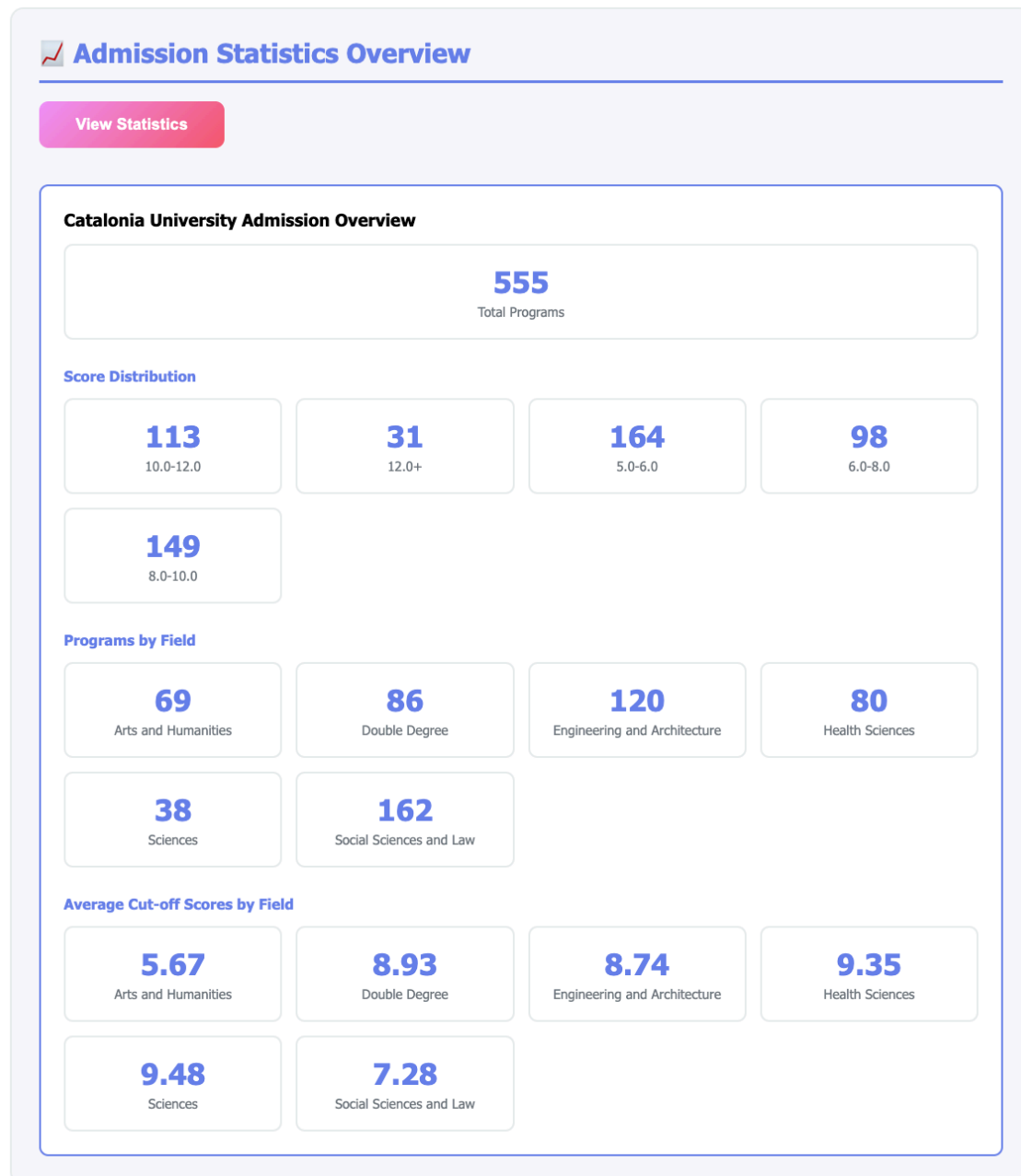
University: UAB Location: Cerdanyola del Vallès Field: Double Degree Cut-off Score: 11.24

Specialization: Social Sciences
- 4. Economia / Estadística (simultaneïtat) (Barcelona)** Safety Double Degree

University: UB / UPC Location: Barcelona Field: Double Degree Cut-off Score: 10.9

Specialization: Social Sciences

At the bottom of the interface, you can access an **overview of statistics**, which summarizes information about all 555 bachelor programs in Catalonia.



4. Conclusion

This project aims to provide a personalized advisory application for high school students in Catalonia, making it possible for them to match their predicted Bachillerato and PAU grades to university degree programs in a simple and interactive way. The current version offers a practical tool for (i) calculating admission scores, (ii) instantly retrieving the top 20 bachelor programs tailored to each student's academic profile and (iii) retrieving general statistics about cut-off scores in Catalonia, with a core focus on usability and clarity.

Looking ahead, there are valuable opportunities for enhancement. An important next step would be integrating a comprehensive search endpoint, allowing users to filter and explore all available degree programs directly from the front-end, not just the top 20 suggestions. This expanded filtering capability would grant students even greater flexibility and empower them to make more informed decisions by accessing detailed information for any degree of interest.